3

Natural history

THIS CHAPTER IS ABOUT KNOWING the variety of the world – about describing and collecting, identifying and classifying, utilising and displaying; it is about the 'notebook' cultures of men and women who loved to 'take note' of their surroundings – not chiefly for meaning, nor necessarily for use, but for the wonder of it or from a compulsion to identify and collect. As already discussed, the term 'natural history' here covers all the things that can be named and collected, not just the animals and plants to which 'naturalists' now are usually devoted. The chapter analyses the history of collecting, describing and displaying – for pride of possession, for intellectual satisfaction, and for commerce and industry.

Natural history was a common term in the seventeenth century, when it had the wide meaning I use in this book. It was the register of facts, the compilation of what was in the world. It contrasted with natural philosophy, which was the account of *causes*, a matter of explanation rather than inventory. Natural philosophy will be discussed further in the next chapter; here I concentrate on how inventories became possible and desirable, and the forms they have taken. We have already seen how natural history could be set against hermeneutics, and 'natural objects' might be 'revealed' by 'removing' the symbolic meanings, the mythology, etymology, etc. But that formulation is problematic; it may suggest that what was removed was 'accretions' whereas it was in fact an integral part of the meaning of a plant or an animal in the Renaissance cosmos. So rather than asking how meanings were stripped away, we do better to ask how 'natural objects' were *created* in the move away from Renaissance worlds.

The tradition of natural objects as bearers of meaning had been nicely epitomised by the cabinets of curiosities which became fashionable from the sixteenth century, and which have recently attracted the attention of historians. The 'cabinets' were meant to be amazing; the collections were of many and varied objects – found or made, each with its own 'story' or message. A horn from a unicorn (or narwhal), a monstrous newborn cat, a stone with the shape of a fish, a rare crystal or jewel, antique coins, or perhaps stone implements; they were objects of significance and of prestige (Findlen, 1994).[1] Such collections were often maintained in high-status sites through to the later eighteenth century; at popular level they continue to our day, in private houses or in 'freak shows' (Altick, 1978; Elsner and Cardinal, 1994). They are entertainments, to be sure, but we do well to remember *that* aspect of *all* displays; it is not by the appeal of zoological taxonomy that the great natural history museums attract so many visitors; and no matter how 'scientific' the exhibits, most visitors will latch on to oddities and remember the museum for the great spindly spider crab that sat in a box at the top of the stairs. Our historical problem is not to explain the continuation of that hermeneutic aspect of collections, but to see why it was partly displaced by more sober ranks of specimens which were meant to be seen, not so much as individually meaningful, but as parts of the ordered arrays of God's creation and of human artifice. Why, by the eighteenth century, were cabinets of curiosities replaced by new collections prized for their 'coverage' of plants or shells, minerals or coins (Hooper-Greenhill, 1992)?

As always in explaining such developments historically, we look for (and to) alternative approaches that could be drawn on and developed – perhaps for subsidiary aspects of culture that might become more prominent; and we look for 'contexts' in which such developments seemed desirable. Here we note two such approaches, which we can call Aristotelianism and the new art; and two such contexts – the exploration of new continents, and the celebration of earthly life and its material riches.

'Historia' and representation

Within Aristotelianism, classification had been a major concern, and so important was this tradition that 'natural history' as it emerged in the seventeenth century might be regarded as a modification of Aristotelianism rather than an alternative to it. The academic botanists of the Renaissance often saw themselves as continuing the work of Theophrastus, the disciple of Aristotle who had focused on plants. Aristotle indeed remained a major resource for 'biologists' well into the nineteenth century; Charles Darwin spoke of Linnaeus and Cuvier as his masters – but 'they were mere pygmies to old Aristotle'. If historians of science tend to disparage the easy naturalism and 'purposiveness' of Aristotle's universe, it is partly because they overlook questions of description and classification in everyday knowledges.

We have argued above that Renaissance scholars tended to subordinate description to meaning. Texts were crucial and natural history was subordinated to natural philosophy. If animal and plant forms were described, it was primarily to show the ways in which the parts functioned in support of the whole, and the ways in which animals were 'self-crafting' – realising in development the purposes inherent in the 'seeds'. Gianna Pomata has argued that part of the seventeenth-century shift to a new form of natural history was an upgrading of description, a readiness to *record* while worrying less about the underlying principles. Doctors, for example, began to publish case histories, as specimens of disease and of their practices. Plants were described 'for their own sakes', not just as remedies (Pomata, 1996).[2] We may link these changes of attitude to the discussions in the previous chapter about market cultures and the celebration of everyday life, about the growing appreciation of crafts and the perception of God as craftsman rather than 'author'. But we can also link them to new techniques of visual representation.

If Aristotelian naturalism was reborn in the universities, art was reborn in the cities of Renaissance Italy, especially in fifteenth-century Florence. It was, and remains, a much more conspicuous feature of their cultural achievement, but one which we as heirs of Romanticism too easily place in opposition to 'science'. We may need to be reminded that the new architecture and naturalistic representation of human bodies were intellectual as well as aesthetic ventures. Central to these arts and sciences was the creation of *perspective*. As Santillana claimed, the 'discovery of perspective, and the related methods of drawing three-dimensional objects to scale, were as necessary for the development of the "descriptive" sciences in a pre-Galilean period as were the telescope and the microscope in the next centuries, and as in photography today' (Santillana, 1959: 33).

From Brunelleschi's experiments in the early fifteenth century, to Masaccio's frescos in the Santa Maria Novella, architects and artists, especially in Florence, struggled to show spatial relations on a plane as they appeared to the eye, e.g. in a camera obscura. In depicting buildings or bodies, the parts were to be related to each other in this perspectival frame. These spatial relationships were central to the new anatomy, best known from the magnificently illustrated volume, the *Fabrica*, published by Vesalius in 1543. As scholarly anatomists pored over editions of Greek texts to present a reborn account of the human body, so artists explored the relationships of the parts and found ways of depicting 'innards' that were as 'lifelike' as the external features more commonly presented in paintings and sculptures (Kemp, 1990; 1997).

One can see it happening. In medieval anatomy texts, all the illustrations are diagrammatic; in Vesalius, all is naturalistic – bodies stripped of skin, or variously splayed out, or skeletons posed in landscapes. In between, *c.*1500, you can find naturalistic bodies with diagrammatic innards – Renaissance outsides, medieval insides. The new pictorial discipline, presenting the world as to the eye, helped open the human body as a land to be explored (Herrlinger, 1970). Where medieval anatomies had been demonstrations of theory, by the seventeenth century they were often seen as part of the new natural history.

Throughout the early modern period, *exploration* was a key to anatomy, and parts of the body were 'seen as' other parts of the world – as fountains and streams, or as branched like plants, etc. The new anatomy paralleled the voyages of exploration then revealing the contents of new continents and shipping them back to the old.

New worlds, new properties and new creators

The exploration of lands unknown to the ancients, and the enormous collections of new plants and animals built up in the capitals of the trading nations, required new forms of inventory. But specimens from the New World came to Europe without the mythology and emblematic significances which were part of the classical tradition for European plants (at least for medicinal and culinary herbs). It is an instructive coincidence that the first botanical texts to include New World plants were also the first to drop the 'human', symbolic aspects, to create natural histories containing 'nothing of the Imagination' (Ashworth, 1990: esp. 322; Grafton, 1992). Likewise for archaeological specimens – remains found in Northern Europe, like New World plants, did not easily fit into written histories derived from ancient Greece and Rome. For Roman coins there was a rich literary context; much less was available for British remains. So these too came to be treated as physical objects which could be listed and catalogued, to form part of a new natural history.

But *why* were these new kinds of materials being imported and/or collected? What created the need for taxonomy? Here we turn to the growth of urban commercial society (especially in Holland), but also to religion (and especially Protestantism, as introduced in the last chapter). We begin our exploration with notes on seventeenth-century Holland, and then move to eighteenth-century England, before returning to questions about the knowledge forms of natural history.

The Dutch case has been beautifully analysed by the art historian Svetlana Alpers in her pioneering book on *The Art of Describing* (Alpers, 1983). She stepped outside a tradition of art history rooted in classical, heroic, narrative art which was meant to be 'read', and she explained Dutch art as celebrating objects and possession. Her critics have argued that she underestimated the narrative, historical and moralising aspect of Dutch painting, but at least, Alpers's shift of emphasis challenges us to analyse forms of depiction and description that often seem so natural (to us) as not to require explanation. (This chapter seeks to do the same for natural history.) Alpers evoked a prosperous urban society, devoted to the arts of peace rather than war, to trade and exploration, and rather liberal in its Protestantism. Rich merchants collected natural objects along with works of craft and art. Paintings were 'pieces of art', well-crafted images that might 'stand in' for the originals (e.g. flower paintings). 'If the theatre was the arena in which the England of Elizabeth most fully represented itself to itself, images played that role for the Dutch' (Alpers, 1983: xxv). These images were 'naturalistic' more than hermeneutic, and the images were everywhere – printed in books, woven into tapestries or linen, painted on to tiles, and framed on walls. Everything was pictured – from insects and flowers to Brazilian natives and the domestic arrangements of Amsterdammers. Maps were central to this culture, not just as tools for navigation, but as representations of the world. The atlas was a Dutch specialism, and so was topography – the art that linked maps with landscapes.

We have already mentioned the crafts of representation in Renaissance Italy; the Dutch could also draw on lenses and the camera obscura – the 'natural' way to produce an image of nature. The image of a town projected in such a camera became part of the world it depicted. The new telescopes and microscopes projected the very distant or the very small so they could be captured in drawings – by 'a Sincere Hand and a Faithful Eye' (Alpers, 1983: 72–3).[3] Clarity and faithfulness of vision was to be treasured, not least by those who needed and could afford the new spectacles. It was in Holland that lens-grinding became a major craft.

Much of the recent writing on the 'scientific revolution' acknowledges Alpers's historical insight. The best recent studies on experimentation in the seventeenth century make clear the relation of these novel manipulations to the wider culture of 'facts and objects' which contemporaries called natural history. Shapin and Schaffer, in their classic study of Robert Boyle and the air pump, see scientific apparatus as making visible the invisible. The pump experiments, to which we return in Chapter 6, manifested the 'spring of the air', rather as telescopes showed the moons of Jupiter or the microscope the 'cells' that made up cork (Shapin and Schaffer, 1985; Shapin, 1994).

But the representation of nature was not only a pleasurable interest and a mark of affluence – it could also be a religious act. 'Anatomising' whether on corpses or on microscopical insects displayed the workmanship of God. This 'moral enquiry' was peculiarly evident in the public dissections of executed criminals, which in Holland were often held in churches. These fashionable events were steeped in moralism: through the corpse of a sinner, the anatomist celebrated God's wondrous designs. Though that sentiment was not new, it took new forms and had a new relation to human creation (Schupbach, 1982).

In most Renaissance world-readings, the craft products of corrupt humans could not compare with products of God, but the Dutch bourgeoisie made the comparison, and their readiness to see God as a craftsman is a key to the possibility and popularity both of 'mechanical philosophy' and the new natural history. We are often reminded that mechanical philosophers liked to see organisms as 'mechanisms' – but that was mostly intellectual fancy. We do better to focus on the ways in which the whole known world could now be used in understanding its obscure realms – whether microscopic, distant or newly discovered. In the new natural history, natural objects and craft objects were on a par – items in an inventory which provided endless scope for analogy. This is crucial, for as historians of art have taught us, all seeing is 'seeing as'. The anatomists who discovered lymphatic vessels saw them as rivulets of water in the landscape of the body, and the form of blood vessels in humans resembled those in plants (and vice versa). You could inject blood vessels with wax and strip away the surrounding tissues, to produce a miniature bush – which might be instructive, or merely ornamental. Indeed, such wax creations were sometimes combined with other bits of anatomy to produce a kind of floral arrangement with moral overtones (Cook, 1996). They celebrated the resemblances of animal and plant parts, and the relationship of nature with wax-modelling. We might call them 'mixed media' and think of Damien Hirst's artwork – the sheep in formalin.

In this culture of possession, natural history was necessarily a matter for the craftsmen and tradesmen who supplied that culture, but so it was also for surgeons, apothecaries and physicians (the university graduates of the medical world). If we skim through Dutch collections of objects, we find the skeletons and monsters for moralising and curiosity; but we also find collections of plant material ranging from the fashionable tulips to exotic spices and herbs, from new kinds of wood for furniture to medicinal plants – perhaps stored in the Delft pottery jars which became symbolic of elite apothecaries. Their shops looked like museums; catalogues of creations became continuous with catalogues of commerce. And by the later seventeenth century, the same was true of London.

When the Royal Society was established there after the Civil War it created a museum, including artefacts, many of them exotic. Like much else in the Society's foundation, the museum was intended to advance the useful arts (Arnold, 1992). In this respect the Society (for a while) continued the endeavour of the puritan Baconian reformers who had pursued the improvement of trades, agriculture and medicine as part of their commitment to a New Jerusalem here on earth. 'Natural history', in my sense and theirs, was central to that enterprise (Webster, 1975); and so were 'ways of doing' in crafts and trades, including 'recipes' – then a common item of cultural and commercial exchange, ranging from cooking to skilled trades, from remedies to tips on the investigation of nature. They stand in a tradition that runs from the 'natural magic' of the Renaissance to the household and trade manuals of the nineteenth century, and to the countless 'how to do it' books (and television programmes) of our day.

Though we know much less about medical practices and commerce than about treatises on physiology, medicine seems to fit this pattern of natural history. The education of apothecaries was central to the development of field botany: apprentices were taken on expeditions to collect 'simples', and for some of them the identification of plants became an end in itself (Allen, 1976). Botanical gardens developed out of herbal gardens, set up by universities or by medical guilds. Chemistry, too, was in part a matter of developing new medicines, in part a matter of improving other trades and crafts, including agriculture. By 1700 in Holland, and by 1750 in Edinburgh, much of this new chemistry, botany and anatomy was included in the education of physicians; the knowledge helped your career as a teacher.

Studies of diseases, too, became more 'natural historical'. 'Case histories' were compiled as contributions to the collective store of observations (and as advertisements of the experience, acuity and effectiveness of particular physicians), and some physicians attempted to characterise epidemics by describing the 'constitution' of the place and time, rather than focusing on the constitutions of the individuals involved. This kind of medicine drew on classical traditions, especially the writings attributed to Hippocrates. Its chief British representative was Thomas Sydenham, a reformer of medicine who moved in the same circles as the physician and philosopher John Locke, and who shared that interest in plants and classification which became seminal for so many eighteenth-century 'savants' (Bynum, 1993; Raven, 1968).

Natures for pedigree people

If seventeenth-century Holland serves as the model for the arts of describing and of making in urban societies, eighteenth-century England serves as a model for the rural; indeed, it may well be our best example of a culture in which the natural-historical was hegemonic. It is no accident that this text is moving from Holland to Britain – so did economic and maritime dominance, and so, later, did the centre of Western medical innovation. From the middle of the eighteenth century, the Scottish universities were notable centres of educated culture – admired by radicals in France, influential in the colonies of North America, and linked by Hanoverian connections to some of the German states and Low Countries. In the Scottish Lowlands and the counties of England, the aristocracy and their imitators among the gentry and merchants built a high culture based on country estates – modelled on classical Greece but revelling in modern knowledge and utilities. Estates were 'improved' to look more like classical landscapes – all the better to be painted. Mountains which had once seemed threatening came to be seen as picturesque or sublime; they were visited, pictured and celebrated in verse. Nature was domesticated and improved, as were fields, plants and farm animals. Cultivated gentlemen collected plants and animals, eggs and snail-shells, minerals and archaeological remains, as well as coins and prints, sculptures and paintings. Such collections advertised wealth, intellect and order; and in as much as they commanded respect, they helped naturalise the social hierarchy (Ritvo, 1987; Thomas, 1983).

Amusingly, British 'history' was then rendered visual and collectable in Granger's system of portraits of notables – a grid of hierarchy against time, a matrix which could be filled with commercial prints of aristocrats, rather as later generations would collect stamps for their albums (Pointon, 1993). Scrapbooks were common and some families bought printed books that could be taken apart so that extra prints (or even specimens) could be inserted. The growing market in descriptive books and engravings mirrored and complemented the collection of specimens and of experiences. Works of God, of man and of art were all superposable – such arrangements might be seen as the dilettante version of the classical episteme, the grid of knowledge described by Foucault. It was in such cultures that formal taxonomies, especially of plants, first came into common use. The debates around them are still instructive, for classification remains a fundamental aspect of old and new knowledges, not least in our 'age of information'.

As eighteenth-century philosophers showed, there were two ways to approach classification. One was to group species with many similar features, and so construct a nesting of similarities. All plants that looked rather like roses could be grouped together as a family; within that large group were smaller ones – for those like strawberries and those like apples, and for those even more like roses. And, at a higher level, the big 'rose family' could be aligned with the daisy family or the umbrella-flowered plants, as opposed to grasses, say. That goal, of a natural system, has remained as a recurrent principle of classifiers. In the twentieth century some taxonomists practise a form of classification called *cladistics*; they use computers to try to establish hierarchies based on maximal numbers of 'characteristics' – all treated as equally important. Some classifications of bacteria work on this basis, which some taxonomists see as uniquely 'objective'. For eighteenth-century philosophers, such a method approximated the mental processes of 'association' which were fundamental to learning (Daudin, 1926; Jacob, 1988).[4]

But such systems were impracticable as guides to identification and arrangement because they depended on many characteristics and were subject to perpetual adjustment as new species were discovered. The alternative was to start with simple characteristics and arrange plants accordingly. Around 1730 the Swede, Carl Linnaeus, devised a wonderfully workable arrangement and procedure. In Holland, at the centre of botanical importing, he came to appreciate the need for classifications that would allow botanists, horticulturalists or apothecaries to know whether or not they were dealing with the same species, whatever the geographical origin of the specimen. To this end Linnaeus invented the binomial naming system and made it stick. (So the common daisy still has a generic and a specific name he gave – *Bellis perennis.*) He linked this mode of naming with his 'sexual system' of plant classification, based on the disturbing, even titillating, discovery that flowers had sexual parts (Frängsmyr, 1984; Stafleu, 1971). In the Linnean classification, the number of male-bits and female-bits found together in flowers served as a God-given principle of order, a divine arrangement which proved its utility and enhanced the pursuit of botany, and of those other realms of nature to which Linnaeus devoted his magnificently orderly brain. It was in the patrician culture of Britain that his legacy was most fully developed and a national botanical society established in his name.

We need to recall the excitement of classification as central to knowing. Classification posed the question of man's place in nature, and his relation to other animals. Classification could be the key to medicine: if only diseases could be properly classified we could more easily recognise them and their resemblances. But here, as elsewhere, intellectual aims could not always be realised; it proved impossible to agree on classifications of disease, mainly because there was no agreement about what constituted a 'disease'. If you examined the relevant text by William Cullen, Scotland's leading physician *c.*1750, the diseases at the twig points of the classification would seem to you but symptoms, for the most part (Cullen, 1816).

But even where one could not achieve a stable classification, one could still describe and compile. The eighteenth century was also the time of encyclopedias, and such collections included technologies and crafts in all their wonderful geographical variety (Gillispie, 1972). One benefit of including artefacts as part of our 'extended natural history' is that we, like the authors then, can see the similarities between craft products and natural species.

Craft products were regional and in some ways they helped to define and characterise the regions. Different regions made different kinds of spades or cloths or houses; indeed, the peasants themselves differed between regions (unlike the 'cosmopolitan' bougeoisie who observed them).[5] Craft and agricultural products were often studied along with animals and plants, minerals and archaeological remains, diseases and the weather – as part of *chorography,* the natural history of *places over time* (Jankovic, 2000). Crafts 'sprang from the soil' like characteristic plants or the vernacular architecture; like plants they were seasonal and affected by the weather. Though we sometimes think of crafts as mechanical, and opposed to biology, this view hardly bears examination. Most crafts converted animal and plant materials into food, clothes, housing or implements, so it is hardly surprising that the products were described and classified like plants and animals.

But we can push the point further, by reflecting on our present knowledges in ways that help us see the past. We now think of many consumer items in terms of 'species' – or 'brands'. Luxury goods – hand crafted, more or less traditional – seem to be understood on 'species' models, while utilitarian (often analytical) characterisations serve for workaday products. We may specify the useful qualities we want from a heating system or an office carpet, but for fine wines or cheese, stylish clothes or racehorses, 'we' demand a quality brand or a prestige breed (Pickstone, 1997). Such possessions were and are the hallmark of social superiority – pedigree breeds for pedigree people.

And if we want clues as to how an eighteenth-century physician could have assessed his patients, then we could draw on the idea of *connoisseurship*, which we understand best from luxury goods or from works of art. Connoisseurs of wines recognise a kind of wine, its origin, vintage and its normal variations; they also know its characteristic faults and 'diseases'. Likewise, to know a patient well was to understand the character, the context, the way of life, the potential and the failings of a person of that kind. Such was the essence of biographical medicine in the classical tradition which saw disease as disturbance of that which was natural to the individual life. To know a patient, as say sanguine or choleric, was to understand his or her natural form of life, and thus to recognise the departure from that nature which a fever, say, might constitute. To treat was to restore the natural condition, by rebalancing the body and its humours, perhaps by bleeding or a change of 'regimen'.

The biographical physician knew to what diseases the patient's nature was liable, and which of these defects might be associated with particular seasons or places. Or to reverse the matter – he also knew the *nature of a place and/or a season*, and what diseases were so associated. Patients had 'constitutions,' but so did seasons and places; medical connoisseurs understood them, as perhaps they still do. However, that natural history of particular places and times could sometimes be 'in tension' with the kind of natural history which worked with 'universal' taxonomies, e.g. of all flowering plants. The former study was central for country parsons and doctors who *lived* the natural history of their own localities, the latter for cosmopolitan travellers and curators who travelled to collect. From the later eighteenth century, plant collecting became a minor profession, part of scientific imperialism.

Natural empires

I have concentrated on the eighteenth century because natural historical understanding was then hegemonic in European cultures of STM. I shall argue in the next chapter that the period 1780–1870 might be called the 'Age of Analysis', because many new disciplines were then constructed which 'went below' the corresponding parts of natural history by 'taking objects apart'. But this does not mean that natural history disappeared, or even that it was weakened. The nineteenth century, one might say, was the great age of 'scientific' museums, a fact which is hidden from many historians of science by their fondness for 'laboratories'. Those who know Cambridge and Oxford may recall that when these country universities first invested in science, in the mid-nineteenth century, it was by building museums, chiefly for natural history (a kind of science compatible with gentlemanly breadth and the liberal arts). Indeed, most nineteenth-century universities maintained museums as part of their teaching and research. It was only in the late twentieth century that some university museums came to be seen chiefly as 'visitor attractions'.

But the greatest investments in museums were by the nineteenth-century nation-states, in their capitals, as expressions of national and imperial power. Paris led the way, when after the Revolution, the former Royal Botanical Garden was developed as a national Museum of Natural History, with professors as curators. Many other former royal collections were turned to public use in comparable ways, including the art collections of the Louvre, and the technical exhibits collected in the

Conservatory of Arts and Manufactures. Under Napoleon, the conquered nations of Europe and North Africa were systematically looted in order to extend the imperial hoards. These collections not only displayed the power of France; by claiming to be definitive collections of the natural order and of great art, they also identified France with science and civilisation. Patronage of museums became a responsibility and a resource of the modern state, and a few devotees gained careers as curator/professors (Hooper-Greenhill, 1992; Pickstone, 1994a).

The British Museum, based on the eighteenth-century collections of Hans Sloane played a similar role in London, along with the botanical gardens at Kew, initially developed as a royal estate. By the mid-nineteenth century, they were seen as collecting places for imperial treasures and as inventories of imperial possessions and resources. The natural history collections were given a huge new building in South Kensington in 1881, well before the government invested in major laboratories. The collections of the (national) geology museum were also moved to South Kensington, alongside the collections of applied arts and of machines and scientific instruments that we will discuss in the next section (Forgan, 1994; Stearn, 1981). Much of the natural history material came from the British imperial possessions. Some of it was collected on government-sponsored expeditions, the most famous being the voyage of HMS *Beagle* in the early 1830s, on which Charles Darwin was the gentleman naturalist. Often, as in this case, the chief purpose was surveying for the improvement of navigation; in mid-century the British navy was unchallenged, so 'research' was a useful stimulus for bright young officers.

Until the end of the century, American expeditions and collections were 'internal' rather than overseas, but it is hard to overestimate the importance of collections, museums and documentation in the formation of the 'scientific-nation'. The most obvious institutions were the national museums in Washington, and the associated agencies for investigating the geology, botany, zoology and ethnology of the continent; but the commercial and industrial cities also had prestige museums, often, as in Britain, established by societies of enthusiasts. The universities developed in the last third of the century built their own collections, not least for teaching agriculture (Dupree, 1957).

From about 1900, all the major European powers (and the USA) invested heavily in colonial development to improve tropical agriculture and decrease the hazard of disease. All such work required infrastructure – of collections, museums and botanical gardens, and these were established in the colonial capitals as well as in the imperial centres. These enterprises remained central to imperial development through the mid-twentieth century, and to the status of postcolonial powers thereafter (Farley, 1991; Sheets-Pyenson, 1989).

Of course, not all the scientific work of museums and expeditions is to be classed as natural history. We shall see in the next chapter that museum professionals asserted their authority by developing analytical approaches, such as comparative anatomy, and by the end of the nineteenth century some of them pursued experimental methods, for example in the new science of ecology. However, even in universities, and *a fortiori* for city and national collections, the basic rationale remained the documentation and display of diversity, for economic, cultural and political reasons. Though historians of biology are fond of arguing that laboratory science *replaced* natural history in the later nineteenth century (Allen, 1978; Maienschein, 1991), this is an exaggeration. 'Laboratory' biology became hegemonic, as we shall discuss later, but taxonomy remained a vital part of the exploration and exploitation of the world, not least of the European empires that flourished in the first half of the twentieth century.

And so it remains, though the empires are now more obviously commercial. The imperial collections, in all their 'genetic diversity', still represent a major resource for industry, agriculture and 'development studies'. In medicine, too, though most research since the early nineteenth century has been analytical, it required huge amounts of *specimens* and of *data* that were often collected as 'natural history' – for medical 'topographies' of foreign places, as simple statistics on death rates, as accounts of epidemics at home and abroad, or as specimens of human diversity in health and disease. Nowadays, molecular biologists often draw on such material to perform their analyses of variation – in humans and in other organisms – and, as we shall discuss in the final chapter, the ownership of such 'natural resources' can be hotly contested.

But even when natural history was being denigrated by exponents of analysis or experimentation, it remained a major component of *popular* science, an important means by which professional biologists could extend their work and influence, and a key to public understandings of 'nature' – as indeed it remains.

Popular natural history

In surveying eighteenth-century natural history we focused on the landed classes, but such studies were also popular in the 'amateur' middle-class societies that were to be found in many provincial towns by the end of the century, both in Britain and in France. Natural history was at least as important as the physical sciences among the 'scientific' elites of the new industrial towns of Britain (Thackray, 1974). Doctors were often naturalists, and the Industrial Revolution spurred a 'flowering' of field natural history among the industrial bourgeoisie looking for rational entertainment in the countryside. By the 1830s, for example, Manchester boasted a middle-class society (and museum) for natural history, as well as more specialist societies for geology and for botany (including horticulture). In the surrounding towns, later in the century, the study of animals, plants and rocks would usually be combined with archaeology and local history – in the local philosophical society or field club. Though the working classes were allowed into these middle-class organisations on special occasions, for the most part they had their own groups, which often met in public houses. There the local experts would put names to plants; in some cases working men specialised in 'difficult' groups such as grasses or mosses (Allen, 1976; Kargon, 1977; Secord, 1994). For some artisans, botany was connected with herbalism or horticulture, for some it offered social connections, and for many it was a way of exercising talents 'beyond their stations in life'.

Class divisions have softened, but natural history societies (general or specialised) continue to the present. From the late nineteenth century they have sometimes served as volunteer workforces for projects launched and guided by academics interested in (analytical) systematics, or in ecology (an analytical/experimental science of vegetation). Nowadays, with the growth of popular ecological consciousness, local natural history societies may take an interest in pollution, and almost certainly they will be interested in 'conservation'; but still their core activity is the identifying and recording of animals and plants in their locality or region. An enormous literature of guidebooks supports such activities, and museums continue to show shells and stuffed birds – though usually now in ecological stage sets rather than filed in classificatory order like library books. From the origins of photography in the mid-nineteenth century, there has been a strong interaction with natural history, and in our age of conservation, 'slides' substitute for specimens. Films, radio and, especially, television have done much to popularise appreciation of 'nature'; indeed, children in Britain now are more familiar with the ways of dinosaurs (Desmond, 1976) and dolphins than with those of blue tits or badgers.

Displays of technology, new and old

We have already noted that 'spotting' and collecting is not confined to natural objects. Over the last two centuries, as before, certain kinds of machinery have been treasured and recorded – railway engines and aeroplanes especially, but also cars, buses and ships. Though sometimes regarded with suspicion when they were new, such machines came to attract affection, and often became part of the 'landscape', almost part of nature. The industrial age has also seen its new 'collectables': postage stamps are the paradigm case, but cards from cigarette packets were once collected, especially by boys, and sports memorabilia (e.g. football match programmes) now play a similar role. In a world of manufactured commodities and aggressive advertising, human products have become more obvious as objects of pursuit and as avenues for identification, collection and display (Elsner and Cardinal, 1994). I once dragged my younger son around Linnaeus's garden, still preserved in Uppsala, and completely failed to interest him in the principles of botanical taxonomy; he was indifferent to the arrangement of the parts of flowers, but when we passed the cycle shop he displayed an astonishing command of the possible arrangements of the parts of mountain-bikes.

From about 1840, *displays* of technology were a feature of industrial cities; they were intended to instruct the middle classes and/or working classes about the range of machinery (and perhaps about its principles). From mid-century, international displays of technology became common – largely as trade fairs in which countries and companies displayed their latest wares and competed for reputations (and orders). The biological mixed with the mechanical; Richard Owen, the cantankerous zoologist from Lancaster who was to head London's Natural History Museum, was a jury chairman for Raw Materials at the Great Exhibition of 1851, and then at the 1855 exhibition of 'Prepared and Preserved Alimentary Substances' held in Paris! (See Desmond, 1994; *Dictionary of National Biography*, Owen; Rupke, 1994.)

Indeed, the 1851 exhibition is nicely illustrative of the many interconnections of mid-Victorian museums culture. The Museum of Practical Geology was completed that year, near Piccadilly. When the Crystal Palace, built to house the Great Exhibition in Hyde Park, was moved to Sydenham, Owen designed huge model dinosaurs as a public attraction. From the profits of 1851, permanent galleries were established in South Kensington to show the best of British 'design' (now the Victoria and Albert Museum) and of British technology and science (the roots of the London Science Museum) (Pointon, 1994). Behind these promotional displays were the masses of manufacturers' catalogues which are invaluable now if you want to reconstruct the material culture of long-gone industries, trades or professions. Technical libraries, too, were a nineteenth-century creation, including patent libraries (and collections of patent models). From that time, the Western world has supported huge and varied resources for 'extended natural history' – from bureaus of standards to type collections of bacteria or viruses. But, of course, many such 'contemporary' collections became 'historical' with the passage of time – and this tension between the novel and the historical has continued to be a feature of (some) science and industrial museums throughout the twentieth century (Butler, 1992).

Since the Second World War, technology museums have often been established primarily to display *old* technology or industrial archaeology. Many of these museums are on industrial heritage sites – displays of local history and social history. They are not concerned with novelty, nor with the analysis of artefacts or instruments; they show the historical natural history of technology – either by displaying ranges of related artefacts, or by putting artefacts into 'reconstructed settings' (Butler, 1992). In Scandinavia similar displays are to be found in 'folk museums' linked to academic 'folk studies'. In Britain and America, 'folk studies' remained a largely amateur pursuit (part of natural history) (Dorson, 1968).

British collectors, from the mid-nineteenth century, preferred archaeology and anthropology – the former concerned largely with pre-Roman Britain, the latter with the British Empire. Both had their amateurs as well as professionals, and in Victorian Britain much of the activity could be regarded as 'natural history' – collections and classification of artefacts, or customs or languages, etc. A marvellous example of the mixture can still be seen in Oxford, at the back of the Victorian (natural history) Museum (promoted by John Ruskin), where the university houses the Pitt-Rivers Collection of historical and anthropological artefacts. These are displayed by function – e.g. devices for making fire or for scaring spirits – an enormous 'junk shop' arranged like a department store, an assemblage from other cultures which wonderfully echoes the material culture of our own (Chapman, 1985).

'Natural history' now

In this chapter I have tried to trace some histories of a way of knowing I call extended natural history. I have linked description, 'biography,' classification and display and related them both to techniques and to social movements. I have described natural history as a foundation for more complex STM, as an aspect of the economy and as a key mode in which western people have related to their worlds, including worlds that are full of technical devices and concepts. In the final chapter of this book I will return to some of these questions, for our present *c.*2000, but here I would underline two features of natural history which we shall see in the intermediate chapters.

First, the *coexistence* of natural history with other ways of knowing, and the continuing contests around their relative importance – even in museum-based taxonomy, an activity sometimes looked down on as largely 'mechanical'. For example, as we noted, some taxonomists (called 'transformed cladists') now like to assess relationships between specimens by counting common and divergent characteristics; we might say they practise a sophisticated form of Enlightenment natural history. Other classifiers may be more mindful of characters that are regarded as 'fundamental' or 'elementary'; we shall note this possibility when discussing classical (analytical) Victorian botany in Chapter 5. However, most taxonomists since the late nineteenth century have seen classification as a way of exploring evolutionary sequences, an approach which some cladists regard as speculative. Evidently, even in its most technical aspects, natural history can still be controversial. But what of its wider roles and principles?

In our customary, narrow sense of natural history, 'conservation of nature' now seems the ruling principle. Where the early moderns *explored*, and our more recent ancestors *classified*, we *conserve*. We worry about loss of habitats in the West and in the Third World, and about loss of species and genetic variation. For most of us now, the protection of nature is less a matter of theology than of aesthetics and the precautionary principle (Bramwell, 1989). Extended natural history, however, is not just about 'nature', whatever that might mean; it also covers all the things which men and women create and use, and at this level the late twentieth century is especially interesting for two reasons. One is the increasing penetration of global commerce and the importance of 'brands'; the other is our 'information systems'.

In this chapter I have stressed the role of commerce – of possession and consumption, and the conceptualisation of craft objects and of luxury goods as quasi-species. I linked such 'species' to 'natural species' and I contrasted them with 'utilitarian products' (which in the next chapter will be shown to be products of analysis). But, so pervasive now are 'brands', and so far are we from the routines of 'traditional' agriculture or any real exposure to 'nature', that the system of references may be reversing. The world which is 'natural' for us is not in the fields, or the natural history museums, or in accounts of the changing seasons in Hampshire villages, but in our product-packed homes and gardens, in the supermarkets and the guides to consumption that flood the television and the magazine racks – not to say the Internet, by which this commercial development interacts with the massive increase in the technical capacity for handling 'information'.

Perhaps our World Wide Web can be compared with the invention of the printed book in the fifteenth century, and of mass publication in the nineteenth. The former transformed the distribution of texts and allowed pictorial representation to become part of the 'descriptive sciences', the latter allowed the popularisation of natural history and the creation of 'domestic manuals', etc. At the end of the twentieth century, computer technology hugely expanded the potential for accumulating and sorting 'information'; where recording methods can be standardised, specimens or catalogue details can now be compared worldwide. 'Virtual' collections, virtual museums, mega-libraries all now beckon. The technology is omnivorous, it minimises distinctions; artworks or wild flowers, details of patents or of bacteria, can all be electronically 'digitised' as information. Yet, presently, this world is wonderfully chaotic – you search by index words, not according to a hierarchical classification – *words* are the key. So, magically perhaps, we return to the Renaissance – to a material world that is ordered by 'words', to the power of (brand) names, and to a system of (commercial) enchantment which penetrates our daily havings and doings as once the claims of religion gave meaning to all earthly things. As we suggested in the last chapter, *dis*enchantment is only half the story.

In such ways our present encourages reflexive exploration of the 'material cultures' and 'information systems' of the past. Into such endeavours this preliminary survey of 'extended natural history' may some day fit. To such questions we will return at the end of the book, after we have discussed the characteristic, hegemonic 'ways' of nineteenth-century science – analysis and experimentalism.

Notes

1 For more on monsters see Daston and Park (1998).
2 I thank the author for sending an English version.
3 The phrase is used by Alpers in the title of her chapter 3, and taken from R. Hooke, *Micrographia*, London, 1665, A2^{v}.
4 For an interesting argument about taxonomies across cultures, see Atran (1990).
5 I owe this point to Marie-Noelle Bourget.